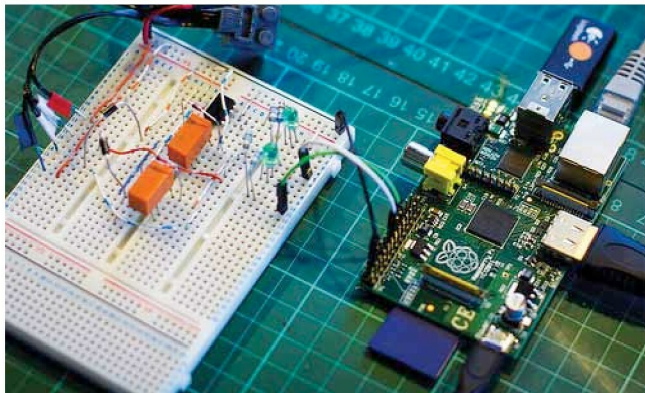




# Control Raspberry Pi's GPIO Pins with PHP

Ever since the tiny credit-card-sized Raspberry Pi appeared in the market, it has caught the imagination of every electronics and computer hobbyist around the world. The powerful Linux OS, combined with its 26 I/O pins, can do many amazing things out-of-the-box. This article is about controlling the GPIO pins.

Besides GPIO (General Purpose Input/Output) pins, the tiny Raspberry Pi has a I2C (Inter-Integrated Circuit) bus, a UART (Universal Asynchronous Receiver/Transmitter Serial) bus and a SPI (Serial Peripheral Interface) bus for connecting several kinds of devices. For the OS installed on the tiny SD card, I prefer Debian Wheezy Linux, for it's a Debian type and easy to install.

To control Raspberry Pi's (or 'RasPi') seven GPIO pins, there are simple Python programs. With a few lines in Python, one can control these pins, making them high or low. Combined with other programs, these pins can do many amazing things. However, here we will do some simple programs like switching some LEDs with these GPIO pins.

Many interactive games can be designed with the help of these GPIO pins. The possibilities are endless, depending on your imagination.

However, since I love PHP, I always try to do the equivalent of most code in PHP—and when I search the Net, I find there is usually a simple way to do this.

## First, the Python way...

Python is already installed on the RasPi; for those who don't know, 'Pi' refers to the Python language. To interact with the GPIO through Python, you have to install the GPIO Python libraries. The latest available is *Rpi.GPIO-0.3.1a.tar.gz*. Download it from <http://raspberrypi-gpio-python.googlecode.com>.